Math 101 — Exam 3 Extra Credit

Part 1 (1 point): Complete all 30 questions on separate paper and show handwritten work. Answers are provided in red; answers alone earn no credit.

Part 2 (2 points): Create and solve 20 similar questions, showing handwritten work. You may ask a friend, tutor, or ChatGPT for ideas or help, but try independently first.

Total extra credit: 3 points.

1. Simplify $\dfrac{x^2+5x+6}{x^2-4}$.

Answer: $\dfrac{x+3}{x-2}$, $x \neq \pm 2$

2. Simplify $\dfrac{x^2-9}{x^2+x-12}$.

Answer: $\dfrac{x+3}{x+4}$, $x \neq 3, -4$

3. Evaluate $16^{-1/2}$.

Answer: $\dfrac{1}{4}$

4. Evaluate $27^{2/3}$.

Answer: 9

5. Rationalize $\dfrac{5}{\sqrt{3}}$.

Answer: $\dfrac{5\sqrt{3}}{3}$

6. Rationalize $\dfrac{2}{3+\sqrt{5}}$.

Answer: $\dfrac{3-\sqrt{5}}{2}$

7. Compute $(7+4i)-(3+9i)$.

Answer: $4-5i$

8. Compute $(2-6i)+(8+3i)$.

Answer: $10-3i$

9. Find the domain: $f(x)=\dfrac{x+1}{x-5}$.

Answer: $(-\infty, 5) \cup (5, \infty)$

10. Find the domain: $g(x)=\dfrac{2x-1}{x^2-9}$.

Answer: $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$

11. Subtract: $\dfrac{3}{x}-\dfrac{2}{x+1}$.

Answer: $\dfrac{x+3}{x(x+1)}$, $x \neq 0, -1$

12. Subtract: $\dfrac{5}{x+2}-\dfrac{1}{x}$.

Answer: $\dfrac{4x-2}{x(x+2)}$, $x \neq 0, -2$

13. Divide: $\dfrac{x^2-9}{x+1} \div \dfrac{x-3}{x+1}$.

Answer: $x+3$, $x \neq -1, 3$

14. Divide: $\dfrac{x^2-4}{x} \div \dfrac{x+2}{x}$.

Answer: $x-2$, $x \neq 0, -2$

15. Divide $2x^3+3x^2-5x+6$ by $x+2$. Give quotient and remainder.

Answer: Quotient $2x^2-x-3$; remainder 12

16. Divide $x^3 - 2x^2 + 4x - 1$ by $x - 1$. Give quotient and remainder.
Answer: Quotient $x^2 - x + 3$; remainder 2

17. Simplify $\sqrt[5]{32x^5}$, for $x > 0$.
Answer: $2x$

18. Simplify $\sqrt[3]{54x^4}$.
Answer: $3x\sqrt[3]{2x}$

19. Solve $\sqrt{x+5} = 4$.
Answer: $x = 11$

20. Solve $\sqrt{2x+1} - 1 = 4$.
Answer: $x = 12$

21. Simplify $\dfrac{\frac{1}{x} + \frac{1}{2}}{\frac{1}{x}}$.
Answer: $\dfrac{x+2}{2}$, $x \neq 0$

22. Simplify $\dfrac{\frac{2}{x} - \frac{1}{3}}{\frac{1}{x}}$.
Answer: $\dfrac{6-x}{3}$, $x \neq 0$

23. Compute $\dfrac{3+i}{1-i}$ in $a + bi$ form.
Answer: $1 + 2i$

24. Compute $\dfrac{4-2i}{1+i}$ in $a + bi$ form.
Answer: $1 - 3i$

25. Simplify $3\sqrt{12} - \sqrt{27}$.
Answer: $3\sqrt{3}$

26. Simplify $2\sqrt{50} + \sqrt{8}$.
Answer: $12\sqrt{2}$

27. One student cleans a room in 6 hours. Together with a second student, they take 4 hours. How long does the second student need alone?
Answer: 12 hours

28. y varies inversely with x. If $y = 8$ when $x = 3$, find y when $x = 6$.
Answer: 4

29. Medicine dosage varies directly with weight. If 150 lb corresponds to 900 mg, what amount corresponds to 175 lb?
Answer: 1050 mg

30. A cyclist travels 30 miles in the time a walker travels 10 miles. The cyclist is 8 mph faster. Find the walker's speed.
Answer: 4 mph